

PROJECT SCOPE MANAGEMENT PLAN

Project: Enhancing Operational Efficiency in a Multi-Specialty Hospital

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1. Scope Management Approach

This document defines the governance, structure, and control mechanisms used to define, validate, manage, and control project scope across the project lifecycle. The approach ensures alignment with business objectives, clinical safety requirements, regulatory compliance, and enterprise project governance standards.

1.1 Scope Governance Principles

- All scope elements must trace to Business Objectives → Requirements → Deliverables → Acceptance
- Scope defined using capability-driven and deliverable-oriented Work Breakdown Structure (WBS)
- Clinical safety and operational continuity are mandatory acceptance dimensions
- Data governance and interoperability standards must be enforced
- Scope baseline cannot change without formal Change Control Board (CCB) approval
- Bidirectional traceability must be maintained at all times

1.2 Scope Performance Metrics

- Requirement coverage: 100%
- Accepted deliverables: $\geq 95\%$
- Scope variance: $\leq \pm 5\%$
- No unresolved critical defects
- Clinical safety validation: Mandatory
- Data integrity validation: Mandatory

1.3 Configuration & Version Control

- Scope baseline version controlled by Business Analyst
- Scope freeze enforced after design approval
- Re-baselining requires Steering Committee approval

2. Roles and Responsibilities

- Project Sponsor – Approves scope baseline and major scope changes
- Steering Committee – Governance, strategic decisions, and escalation
- Project Manager – Owns scope delivery and control
- Business Analyst – Maintains traceability and scope alignment
- IT Lead – Ensures technical and integration delivery
- QA Lead – Validates quality, safety, and acceptance
- Medical Director – Ensures clinical safety and care continuity
- Nursing Operations Lead – Validates workflow practicality
- Compliance & Regulatory Officer – Ensures legal and regulatory alignment
- Data Governance Owner – Ensures data integrity and ownership
- Integration Architect – Ensures system interoperability

3. Scope Definition

Scope is defined as the establishment of an integrated hospital operational coordination capability that improves patient flow, resource utilization, and operational visibility while maintaining clinical safety, regulatory compliance, and uninterrupted care delivery.

3.1 Definition of Done

- All functional and non-functional requirements satisfied
- Clinical workflow validated and safe
- No critical or high defects
- Data synchronization and integrity verified
- Performance validated under peak load
- Compliance and regulatory validation completed
- Operational teams confirm production readiness

4. Project Scope Statement

4.1 Project Objectives

- Reduce average patient waiting time by $\geq 30\%$
- Achieve $\geq 95\%$ successful digital bookings with zero double booking
- Optimize operational resource utilization to 70–85%
- Enable real-time hospital operational visibility
- Improve patient satisfaction by $\geq 20\%$
- Maintain $\geq 99.5\%$ system uptime with regulatory compliance

4.2 In-Scope Capabilities

- Enterprise appointment coordination and scheduling
- Real-time operational resource visibility and utilization management
- Interoperability across HIS, billing, pharmacy, and medical records
- Operational analytics and performance monitoring
- Secure and compliant healthcare data management
- Clinical workflow validation and downtime continuity procedures
- Organizational adoption, training, and change enablement

4.3 Out-of-Scope

- Hospital construction or infrastructure redesign
- Medical equipment replacement
- Clinical staff hiring
- Full legacy HIS replacement
- AI predictive optimization
- Mobile patient applications

4.4 Constraints

- Limited IT budget requiring phased implementation
- Fixed timeline requiring scope prioritization
- Regulatory compliance constraints
- Legacy system dependency
- Staff availability for training and transition

4.5 Assumptions

- Stakeholder availability for validation
- Data quality sufficient for integration
- Operational staff adopt new workflows
- Leadership supports organizational change
- No major regulatory changes during project execution

5. Work Breakdown Structure (WBS)

5.1 WBS Overview

The Work Breakdown Structure (WBS) defines the total project scope using a deliverable-oriented decomposition aligned with enterprise project governance and healthcare operational safety. The WBS follows the 100% rule, ensuring all scope is fully represented, measurable, and traceable to business objectives, requirements, and deliverables.

5.2 WBS Hierarchy

- 1.0 Project Governance & Control – Governed and compliant project execution
 - 1.1 Project Planning & Kickoff – Approved Project Management Plan
 - 1.2 Risk, Quality & Compliance Governance – Active risk and quality framework
- 2.0 Requirements & Scope Definition – Validated and traceable scope
 - 2.1 Stakeholder & Clinical Requirement Validation – Approved Requirements
 - 2.2 RTM & Acceptance Model – Full bidirectional traceability
- 3.0 Appointment Coordination Capability – Safe and synchronized scheduling
 - 3.1 Booking & Availability Engine – Functional booking module
 - 3.2 Scheduling Integrity & Conflict Prevention – Validated scheduling logic
 - 3.3 Clinical Workflow Safety Validation – Safe operational scheduling
- 4.0 Resource Visibility & Utilization Management – Real-time resource intelligence
 - 4.1 Resource Tracking Engine – Doctors, rooms, and equipment tracking
 - 4.2 Utilization & Throughput Analytics – Optimization insights
- 5.0 Patient Communication & Notification – Automated operational communication
 - 5.1 Reminder Automation – Appointment reminders
 - 5.2 Operational Alerting – Delay and change notifications
- 6.0 HIS Integration & Data Governance – Safe and reliable interoperability
 - 6.1 Core Integration Layer – HIS / Billing / EMR / Pharmacy synchronization
 - 6.2 Data Integrity & Consistency – Validated synchronized data
 - 6.3 Interoperability & Standards Compliance – Data governance and standards
- 7.0 Operational Analytics & Performance Intelligence – Measurable decision capability
 - 7.1 KPI Intelligence Layer – Wait time, utilization, and booking metrics
 - 7.2 Operational Reporting – Performance monitoring
- 8.0 Security, Privacy & System Reliability – Secure and compliant system
 - 8.1 Identity, Access & Privacy Controls – Role-based and data protection
 - 8.2 Performance, Load & Failover Validation – Peak-safe system
 - 8.3 Regulatory Compliance Validation – Healthcare compliance verified
- 9.0 Change Enablement & Clinical Adoption – Safe organizational adoption
 - 9.1 Clinical & Operational Training – Users trained
 - 9.2 UAT & Operational Readiness – Adoption validated
 - 9.3 Adoption & Behavioral Monitoring – Sustained usage
- 10.0 Deployment, Stabilization & Operational Transition – Stable production system

- 10.1 Controlled Production Deployment – System live
- 10.2 Post-Go-Live Stabilization – Stable operations
- 10.3 Downtime & Contingency Workflow Validation – Care continuity ensured

5.3 WBS Dictionary

The detailed WBS Dictionary, including work package descriptions, acceptance criteria, owners, dependencies, and risks, is maintained as a controlled appendix artifact aligned with the Scope Baseline and RTM.

6. Scope Validation

- Requirement walkthrough and traceability validation
- Clinical workflow and safety validation
- Data integrity and interoperability validation
- Performance testing under peak load
- Stakeholder and operational sign-off

7. Scope Control

- All scope changes require formal Change Request submission
- Impact analysis must include scope, cost, time, risk, and patient safety
- Change Control Board approval mandatory
- Scope baseline updated after approved changes
- Unauthorized scope additions rejected

8. Scope Traceability

All scope elements must maintain full traceability: Objective → Requirement → Deliverable → Acceptance.

Reverse traceability is mandatory and maintained by the Business Analyst.

9. Scope Baseline

The scope baseline includes the approved Scope Statement, Requirements Traceability Matrix (RTM), and WBS.

Baseline is version-controlled and cannot change without Steering Committee approval.